

## Lab 5 - Week 4

## VGA using SRAM

Design, synthesize, and simulate a Video Graphics Controller (VGA Controller) using SRAM.

**Design Code**

top\_tb.v × top.v × video\_controller.v × address\_generator.v × sram.v × top\_tb\_behav.wcfg ×

D:/SSUET/INSPIRE IC DESIGN/RTL/vga/video\_controller.v

Q

```
23 module video_controller(  
24  
25     input clk,  
26     input reset,  
27  
28     output reg hsync,  
29     output reg vsync,  
30  
31     output reg [9:0] h_count,  
32     output reg [9:0] v_count,  
33  
34     output video_on  
35  
36 );  
37  
38 parameter H_VISIBLE=640;  
39 parameter H_FRONT=16;  
40 parameter H_SYNC=96;  
41 parameter H_BACK=48;  
42 parameter H_TOTAL=800;  
43  
44 parameter V_VISIBLE=480;  
45 parameter V_FRONT=10;  
46 parameter V_SYNC=2;  
47 parameter V_BACK=33;  
48 parameter V_TOTAL=525;  
49  
50 always @(posedge clk or posedge reset)  
51  
52 begin  
53  
54 if(reset)  
55
```

SIMULATION - Behavioral Simulation - Functional - sim\_1 - top\_tb

top\_tb.v x top.v x video\_controller.v x address\_generator.v x sram.v x top\_tb\_behav.wcfg x

D:/SSUET/INSPIRE IC DESIGN/RTL/vga/video\_controller.v

Scope Sources

```

56 begin
57   h_count<=0;
58   v_count<=0;
59 end
60
61 else
62
63   begin
64
65     if(h_count==H_TOTAL-1)
66
67       begin
68
69         h_count<=0;
70
71         if(v_count==V_TOTAL-1)
72           v_count<=0;
73
74         else
75           v_count<=v_count+1;
76
77       end
78
79       else
80         h_count<=h_count+1;
81
82       end
83
84     end
85
86   always @(*)
87
88   begin
89
90     hsync=~((h_count>=656)&&(h_count<752));
91
92     vsync=~((v_count>=490)&&(v_count<492));
93
94   end
95
96   assign video_on=(h_count<640)&&(v_count<480);
97
98   endmodule
99
100

```

SIMULATION - Behavioral Simulation - Functional - sim\_1 - top\_tb

top\_tb.v x top.v x video\_controller.v x address\_generator.v x sram.v x top\_tb\_behav.wcfg x

D:\SSUET\INSPIRE IC DESIGN\RTL\vga\address\_generator.v

Scope Sources

```
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module address_generator(
24
25     input clk,
26     input [9:0] h_count,
27     input [9:0] v_count,
28
29     output reg [18:0] address
30
31 );
32
33 always @(posedge clk)
34 begin
35
36     if(h_count < 640 && v_count < 480)
37         address <= v_count*640 + h_count;
38     else
39         address <= 0;
40
41 end
42 endmodule
43
```

SIMULATION - Behavioral Simulation - Functional - sim\_1 - top\_tb

top\_tb.v x top.v x video\_controller.v x address\_generator.v x sram.v x top\_tb\_behav.wcfg x

D:/SSUET/INSPIRE IC DESIGN/RTL/vga/top.v

Q [Save] [Undo] [Redo] [Cut] [Copy] [Paste] [Delete] [Comment] [Lightbulb]

```
23 module top(  
24  
25  
26     input clk,  
27     input reset,  
28  
29     output hsync,  
30     output vsync,  
31  
32     output [7:0] red,  
33     output [7:0] green,  
34     output [7:0] blue  
35  
36 );  
37  
38 wire [9:0] h_count;  
39 wire [9:0] v_count;  
40  
41 wire video_on;  
42  
43 wire [18:0] address;  
44  
45 wire [7:0] pixel;  
46  
47 video_controller VC(  
48  
49     .clk(clk),  
50     .reset(reset),  
51  
52     .hsync(hsync),  
53     .vsync(vsync),  
54  
55     .h_count(h_count),
```

SIMULATION - Behavioral Simulation - Functional - sim\_1 - top\_tb

top\_tb.v x top.v x video\_controller.v x address\_generator.v x sram.v x top\_tb\_behav.wcfg x

D:/SSUET/INSPIRE IC DESIGN/RTL/vga/top.v

Q [Save] [Undo] [Redo] [Cut] [Copy] [Paste] [Delete] [Comment] [Lightbulb]

```
56     .v_count(v_count),
57
58     .video_on(video_on)
59
60 );
61
62 address_generator AG(
63
64     .clk(clk),
65
66     .h_count(h_count),
67     .v_count(v_count),
68
69     .address(address)
70
71 );
72
73 sram MEM(
74
75     .clk(clk),
76
77     .we(1'b0),
78
79     .address(address),
80
81     .data_in(8'd0),
82
83     .data_out(pixel)
84
85 );
86
87 assign red    = video_on ? pixel : 8'd0;
88 assign green  = video_on ? pixel : 8'd0;
89
90
91 assign green = video_on ? pixel : 8'd0;
92 assign blue  = video_on ? pixel : 8'd0;
93
94 endmodule
```

SIMULATION - Behavioral Simulation - Functional - sim\_1 - top\_tb

top\_tb.v x top.v x video\_controller.v x address\_generator.v x sram.v x top\_tb\_behav.wcfg x

D:/SSUET/INSPIRE IC DESIGN/RTL/vga/sram.v

Q [Save] [Undo] [Redo] [Cut] [Copy] [Paste] [Delete] [Find] [Help]

```
23 module sram(  
24  
25     input clk,  
26     input we,  
27     input [18:0] address,  
28     input [7:0] data_in,  
29  
30     output reg [7:0] data_out  
31  
32 );  
33  
34     reg [7:0] memory [0:307199];  
35  
36     integer i;  
37  
38     initial  
39     begin  
40  
41         for(i=0;i<307200;i=i+1)  
42             memory[i]=i%256;  
43  
44     end  
45  
46     always @(posedge clk)  
47     begin  
48  
49         if(we)  
50             memory[address] <= data_in;  
51  
52         data_out <= memory[address];  
53  
54     end  
55
```

## Testbench

**SIMULATION - Behavioral Simulation - Functional - sim\_1 - top\_tb**

top\_tb.v x top.v x video\_controller.v x address\_generator.v x sram.v x top\_tb.v

D:/SSUET/INSPIRE IC DESIGN/RTL/vga/top\_tb.v

Q [Icons]

```
23 module top_tb;
24
25     reg clk;
26     reg reset;
27
28     wire hsync;
29     wire vsync;
30
31     wire [7:0] red;
32     wire [7:0] green;
33     wire [7:0] blue;
34
35     top DUT(
36
37         .clk(clk),
38         .reset(reset),
39
40         .hsync(hsync),
41         .vsync(vsync),
42
43         .red(red),
44         .green(green),
45         .blue(blue)
46
47     );
48
49     initial
50
51     begin
52
53         clk=0;
54
55         forever #20 clk=~clk;
```

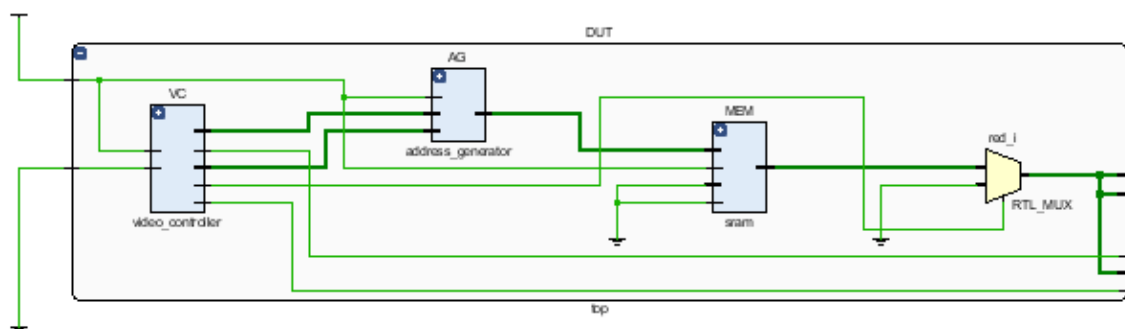
SIMULATION - Behavioral Simulation - Functional - sim\_1 - top\_tb

top\_tb.v x top.v x video\_controller.v x address\_generator.v x sram.v x top\_tb\_behav.wcfg x

D:/SSUET/INSPIRE IC DESIGN/RTL/vga/top\_tb.v

```
57 end
58
59 initial
60 begin
61   reset=1;
62   #100;
63   reset=0;
64   #1000000;
65   $finish;
66 end
67
68 initial
69 begin
70   $monitor("Time=%0t HSYNC=%b VSYNC=%b R=%d G=%d B=%d",
71     $time,
72     hsync,
73     vsync,
74     red,
75     green,
76     blue);
77 end
78
79 endmodule
```

## Schematic Diagram





## Waveform

